

COMS30035, Machine learning:

Unit organisation

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Unit github page

- ▶ Most teaching materials and information are accessed via the unit's github page: <https://uob-coms30035.github.io/>
- ▶ This also gives you the unit schedule.
- ▶ There is also a Blackboard page which has quizzes.
- ▶ At some point we will also be using gitlab.

Teaching staff

- ▶ The only lecturer on this unit is: [James Cussens](#)
- ▶ The following people are GTAs on the unit: Marek Janiec, Russell Summers, Gonzalo Cancio-Donlebun Fernandez, Abby Morris and Tomos Sherlock
- ▶ Both a lecturer and GTAs are present at drop-in sessions and labs.

Teams

- ▶ There is a Teams group “Machine Learning (Teaching Unit) 2026” for this unit. It has 2 public channels:
 - General** Read-only for students. (We might use this for announcements, but we can also use Blackboard for that.)
 - Open** Students can post here. Feel free to use it as you wish; for example to ask questions.
- ▶ At some point I will probably set up a Team group just for students doing this unit as a MAJOR.

Teaching sessions

- ▶ Lectures are 0900 Monday and 1100 Tuesday.
- ▶ There is one 3-hour lab each week starting at 1300 on Thursday.
- ▶ Drop-in session on Thursdays (1700).
- ▶ During week 6 there is no teaching and during the coursework period (weeks 9-11) there is no teaching apart from coursework support sessions (1400-1600 Thursdays).

Textbooks

- ▶ Bishop, C. M., Pattern recognition and machine learning (2006). Available for free [here](#).
- ▶ Murphy, K., Probabilistic Machine Learning: An Introduction (2022). This book is also freely available [here](#).
- ▶ Bishop is used more than Murphy.
- ▶ At the end of each set of lecture slides there will be a pointer stating where to find the relevant material in Bishop and/or Murphy.
- ▶ There will also (typically) be pointers to some problems taken from these textbooks.
- ▶ You can get help with these at drop-in sessions. It's fine to get help with a problem from an earlier week.

Quizzes and problems

- ▶ There are both Blackboard quizzes ('Unit Information and Resources' → 'Quizzes' in *Machine Learning (Teaching Unit) 2026* Blackboard site) and problems associated with the lecture material.
- ▶ Pointers to both are included at the end of the slides for the relevant lecture.
- ▶ These are **formative** not **summative**, i.e. they do not count towards your mark for this unit.

Drop-in sessions

- ▶ You are not obliged to attend any drop-in sessions.
- ▶ You can get help with the quizzes and problems by coming to a drop-in session.
- ▶ But you can also ask us about anything at the drop-in sessions, e.g. further explanation of lecture material, pointers to additional reading, etc.

Assessment

- ▶ If you are doing this unit as a MAJOR (COMS30083), then you are assessed by a mid-term in-class test in week 6 (30%) and coursework (70%) done during weeks 9-11.
- ▶ Coursework is released at 1500 on Friday 13 November. Submission deadline is 1300 on Thursday 3 December.
- ▶ If you are doing this unit as a MINOR (COMS30081), then you are assessed by answering Machine Learning questions (equivalent to a 1-hour exam) in the *Topics in Computer Science* December exam. This is a closed-book exam.

Mid-term test

- ▶ The mid-term test will be a pen-and-paper exam at 1300-1400 on Wednesday 28 October. We will use room MVB 2.11.
- ▶ It will be closed book and you are not permitted to bring notes into the exam.
- ▶ Evidently only material from Weeks 1–5 will be examined.

Revision materials

- ▶ All past assessments (MINOR exam, MAJOR in-class test and MAJOR coursework) are available via the “Machine Learning (Teaching Unit) 2026” team. They are in the ‘Past assessments’ folder.
- ▶ In addition, there is a ‘mock’ MINOR exam and a ‘mock’ MAJOR in-class test, which was supplied to students in 24/25 (in the absence of appropriate past exams).
- ▶ For the MINOR exams and MAJOR in-class tests there is a version with answers and a version without.
- ▶ For the coursework cohort-level feedback is provided (not the individual feedback of course!).

Cribsheet

- ▶ A 'cribsheet' is available from the unit github page (in the Unit Materials section).
- ▶ This is a PDF which contains a list of things you need to know/understand.
- ▶ Occasionally, things you **don't need to know** are given.

Labs

- ▶ During lab sessions you do machine learning exercises.
- ▶ You can use either the machines in the room or your own machine.
- ▶ The machines in the room have the necessary ML software installed.
- ▶ You need to take certain steps to access that software: see the section on Lab Work on the [unit github page](#).
- ▶ If you choose to use your own machine you will have to install the necessary software yourself. This is entirely doable and we're happy to attempt to help you if you hit installation problems, but we can't guarantee to be able to fix all such problems (particularly if you choose to use Windows).
- ▶ Labs are the most important part of the unit since there you actually **do** machine learning, rather than merely hear about it.

The role of labs

- ▶ The coursework will require you to apply the ML software we have been using in labs to machine learning problems. So doing lab work leads directly into doing coursework.
- ▶ Clearly, lab attendance is less crucial for those of you who will not be doing coursework.
- ▶ But, apart from the sheer joy of learning, I encourage students doing this unit as a MINOR to attend labs.
- ▶ Actively using, say, a Support Vector Machine method to learn a classifier, is more educational than passively having them explained to you.

Lab materials

- ▶ The materials for each lab are contained in a github repo:
https://github.com/uob-COMS30035/lab_sheets_public.
- ▶ For each lab you are presented with a Jupyter notebook which contains ML exercises for you to do. You complete each exercise by adding the necessary Python code to the notebook.
- ▶ There are links to these notebooks in the schedule on the unit's github page.
- ▶ Most labs require files (e.g. data, utility Python scripts) in addition to the notebook.
- ▶ The easiest way to ensure you have everything you need for each lab is to simply clone the git repo.
- ▶ (You can update your local git repo of the labs with a `git pull` before each lab, just to be sure you have any updates.)

Answers to formative material

- ▶ Answers to lab exercises and any problems **not** from Bishop or Murphy will be released after the relevant lab.
- ▶ The problems selected from Bishop and Murphy have solutions available from the web sites associated with the book.
- ▶ Quiz answers are not 'released' (they are multiple-choice). The quizzes are set up **not** to give you the correct answer if you give an incorrect answer. This is deliberate!

Python

- ▶ All your coding on this unit will be done in Python.
- ▶ This code will call ML library methods supplied by packages such as scikit-learn, pytorch and PyMC.
- ▶ (Note that Python is mostly used as a wrapper; the native code called by Python was typically compiled from C/C++/FORTRAN source.)
- ▶ Your code will either be a regular script or in a JupyterLab notebook.
- ▶ Python coding, per se, is not taught on this unit.